

Math727 – Numerical Analysis II

Homework 1

1. Let us consider the BVP

$$\begin{aligned} -u''(x) + p(x)u'(x) + r(x)u(x) &= \phi(x), & x \in [a, b] \\ u(a) &= \alpha, & u(b) = \beta, \end{aligned}$$

for $\alpha, \beta \in \mathbb{R}$ and functions $p, r, \phi \in C[a, b]$ with $r(x) \geq 0$ for $x \in [a, b]$. The approximation of the derivatives u' and u'' by central difference schemes of first and second order on an equidistant grid $x_k = a + k(b - a)/N$ for $k = 1, 2, \dots, N - 1$ leads to a linear system of equations $\mathbf{A}\mathbf{u} = \mathbf{b}$ for $\mathbf{u} = (u_1, u_2, \dots, u_{N-1})^T \in \mathbb{R}^{N-1}$ for a certain matrix $\mathbf{A} \in \mathbb{R}^{(N-1) \times (N-1)}$ and a certain vector $\mathbf{b} \in \mathbb{R}^{N-1}$ and with the approximations $u_k \approx u(x_k)$. Compute $\mathbf{A}$ and $\mathbf{b}$ and show that the system of equations is uniquely solvable for sufficiently small h .

2. For a function $\phi \in C[0, 1]$ consider the BVP

$$u''(x) = \phi(x), \quad u(0) = u(1) = 0. \quad (1)$$

- (a) Show that the solution of the above equation (1) can be written in the form

$$u(x) = \int_0^1 G(x, \xi) \phi(\xi) d\xi$$

where the Green function

$$G(x, \xi) = \begin{cases} \xi(x - 1), & \text{if } \xi \leq x, \\ x(\xi - 1), & \text{otherwise.} \end{cases}$$

- (b) Let the functions u and $u + \Delta u$ be solutions to the BVP (1) and its perturbed version

$$(u + \Delta u)'' = \phi + \Delta \phi, \quad (u + \Delta u)(0) = (u + \Delta u)(1) = 0. \quad (2)$$

respectively, where $\Delta \phi \in C[0, 1]$, and $|\Delta \phi| \leq \epsilon$ for $x \in [0, 1]$. Show that

$$|\Delta u| \leq \epsilon x(1 - x)/2.$$

- (c) The difference method using central differences of second order yields as a solution of a linear system of equations $\mathbf{A}_o \mathbf{u} = \mathbf{b}$ the approximations u_j for $u(x_j)$ with $x_j = j/N, j = 1, 2, \dots, N - 1$. For the perturbed variants

$$\mathbf{A}_o(\mathbf{u} + \Delta \mathbf{u}) = \mathbf{b} + \Delta \mathbf{b}, \quad \Delta \mathbf{b} \in \mathbb{R}^{N-1}, \quad \|\Delta \mathbf{b}\| \leq \epsilon,$$

show that

$$|\Delta u_j| \leq \frac{\epsilon}{2} x_j(1 - x_j), \quad j = 1, 2, \dots, N - 1.$$

3. **(Programming Exercise)** Solve the BVP

$$u''(x) + 6x(1-x)u'(x) + u(x)^2 = x^4 + 10x^3 - 17x^2 + 6x - 2, \quad x \in [0, 1]$$

$$u(0) = u(1) = 0,$$

numerically using the simple shooting method. To find the root use the Newton's method first with the initial values $s^{(0)} = 1$ and then with $s^{(0)} = 20$. The respective initial value problems shall be solved using the explicit Euler method with step size $h = 1/30$. Provide a table of approximations u_k for the grid points $x_k = kh$, $k = 0, 1, \dots, 30$ and draw the corresponding figure for each case. Also submit your codes in paper. You could use Language such as C/C++, Fortran or Matlab.